

Section 8.1 Exercise Answers

1. $B = 68^\circ, b = 27.2, c = 27.4$
2. $B = 72^\circ, a = 12.0, b = 19.9$
3. $C = 20^\circ, b = 4.5, c = 1.6$
4. $A = 55^\circ, a = 8.2, c = 5.0$
5. $b = 3.78$
6. $a = 17.58$
7. $c = 13.70$
8. No triangle, no solutions exist.
9. One triangle, $A = 50.3^\circ, B = 16.7^\circ, a = 26.7$
10. One triangle, $B = 40.6^\circ, A = 59.4^\circ, a = 4.6$
11. Two Triangles, $C = 54.3^\circ, B = 90.7^\circ, b = 20.9$ or $C = 125.7^\circ, B = 19.3^\circ, b = 6.9$
12. One Triangle, $a = 14.3^\circ, C = 140.7^\circ, c = 52.5$
13. Two Triangles, $B = 75.7^\circ, C = 61.3^\circ, b = 9.9$ or $B = 18.3^\circ, C = 118.7^\circ, b = 3.2$
14. Two Triangles, $A = 71.6^\circ, C = 84.4^\circ, c = 7.3$ or $A = 108.4^\circ, C = 47.6^\circ, c = 5.4$
15. Two Triangles, $B = 26.8^\circ, A = 143.2^\circ, a = 17.2$ or $B = 153.2^\circ, A = 16.8^\circ, a = 8.3$
16. Two Triangles, $A = 36.9^\circ, B = 115.1^\circ, b = 3.5$ or $A = 143.1^\circ, B = 8.9^\circ, b = 0.6$
17. No triangle, no solutions exist.
18. No triangle, no solutions exist.
19. $A = 47.8^\circ$ or $A = 132.2^\circ$
20. $B = 69.3^\circ$ or $B = 110.7^\circ$
21. 12.3
22. 4.4
23. 12.2
24. 12.3
25. 16.0
26. 11.0
27. One triangle, $x = 29.7^\circ$
28. One triangle, $x = 26.0^\circ$
29. Two triangles, $x = 76.9^\circ$ or $x = 103.1^\circ$
30. One triangle, $x = 52.8^\circ$
31. One triangle, $x = 110.6^\circ$
32. No Solution
33. One triangle, $C = 47.6^\circ, A = 39.4^\circ, a = 20.7$